

Category Theory and Functional Programming

The Curry-Howard Correspondence

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Preliminaries

Textbook

S. Abramsky and N. Tzevelekos (2011). Introduction to Categories and Categorical Logic.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Outline

Typed Lambda-Calculus

References

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Typed Lambda-Calculus

Definition

A **typed λ -calculus** with function types, product types, unit type and empty type, is defined by:

(i) Types

$\sigma, \tau ::= b$	(base types)
$\sigma \rightarrow \tau$	(function types)
$\sigma \times \tau$	(product types)
1	(unit type)
0	(empty type)

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(ii) Terms

$M, N ::= x$	(variable)
$\lambda x.M$	(λ -abstraction)
$M N$	(application)
$\langle M, N \rangle$	(pair)
$\pi_1 M \mid \pi_2 M$	(projections)
$\langle \rangle$	(unit)
absurd M	(absurd)

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(iv) Conversion rules

$$\lambda x.M = \lambda y.M[x := y] \quad (\alpha\text{-conversion})$$

$$(\lambda x.M) N = M[x := N] \quad (\beta\text{-conversion})$$

$$\lambda x.M x = M, \text{ with } x \text{ not in } M \quad (\eta\text{-conversion})$$

$$\pi_i \langle M_1, M_2 \rangle = M_i$$

$$M = \langle \pi_1 M, \pi_2 M \rangle$$

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(iii) Contexts

$\Gamma ::= \emptyset$

(empty context)

| $\Gamma, x : \sigma$, with x not in Γ

(context extension)

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(v) Typing rules

- Hypothesis

$$\frac{}{\Gamma, x : \sigma \vdash x : \sigma} \text{hyp} \quad (\text{if } x : \sigma \in \Gamma)$$

- Function types

$$\frac{\Gamma, x : \sigma \vdash M : \tau}{\Gamma \vdash \lambda x. M : \sigma \rightarrow \tau} \rightarrow\text{I}$$

$$\frac{\Gamma \vdash M : \sigma \rightarrow \tau \quad \Gamma \vdash N : \sigma}{\Gamma \vdash M N : \tau} \rightarrow\text{E}$$

- Product types

$$\frac{\Gamma \vdash M : \sigma \quad \Gamma \vdash N : \tau}{\Gamma \vdash \langle M, N \rangle : \sigma \times \tau} \times\text{I}$$

$$\frac{\Gamma \vdash M : \sigma_1 \times \sigma_2}{\Gamma \vdash \pi_i M : \sigma_i} \times\text{E}$$

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Typed Lambda-Calculus

(v) Typing rules

- Unit type

$$\frac{}{\Gamma \vdash \langle \rangle : \mathbf{1}} \text{1I}$$

- Empty type

$$\frac{\Gamma \vdash M : \mathbf{0}}{\Gamma \vdash \text{absurd } M : \sigma} \text{0E}$$

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Typed Lambda-Calculus

Category associated to the typed λ -calculus

The category **TLC** is defined by:

- (i) Objects: Types
- (ii) Arrows: There is an arrow $\sigma \xrightarrow{M} \tau$ iff $M : \sigma \rightarrow \tau$.

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(iii) Composition

If $M : \rho \rightarrow \sigma$ and $N : \sigma \rightarrow \tau$ then

$$M \circ N : \rho \rightarrow \tau$$

$$M \circ N := \lambda x. M (N x)$$

(iv) Identity arrows

$$\text{id}_\sigma : \sigma \rightarrow \sigma$$

$$\text{id}_\sigma := \lambda x. x, \text{ with } x : \sigma$$

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(v) Unit axiom

Let $M : \sigma \rightarrow \tau$, then

$$\begin{aligned} M \circ \text{id}_\sigma &= \lambda x. M(\text{id}_\sigma x) \\ &= \lambda x. M((\lambda y. y) x) \\ &= \lambda x. M x \\ &= M \end{aligned}$$

$$\begin{aligned} \text{id}_\tau \circ M &= \lambda x. \text{id}_\tau (M x) \\ &= \lambda x. (\lambda y. y) (M x) \\ &= \lambda x. M x \\ &= M \end{aligned}$$

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(vi) Associativity axiom

Let $M : \sigma_1 \rightarrow \sigma_2$, $N : \sigma_2 \rightarrow \sigma_3$ and $P : \sigma_3 \rightarrow \sigma_4$. Then

$$\begin{aligned} M \circ (N \circ P) &: \sigma_1 \rightarrow \sigma_4 \\ M \circ (N \circ P) &= \lambda x. M ((N \circ P) x) \\ &= \lambda x. M ((\lambda y. N (P y)) x) \\ &= \lambda x. M (N (P x)) \\ &= \lambda x. (\lambda y. M (N y)) (P x) \\ &= \lambda x. (M \circ N) (P x) \\ &= (M \circ N) \circ P \end{aligned}$$

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(vii) Initial object: Type 0

$$i_\sigma : 0 \rightarrow \sigma$$

$$i_\sigma := \text{absurd}$$

(viii) Terminal object: Type 1

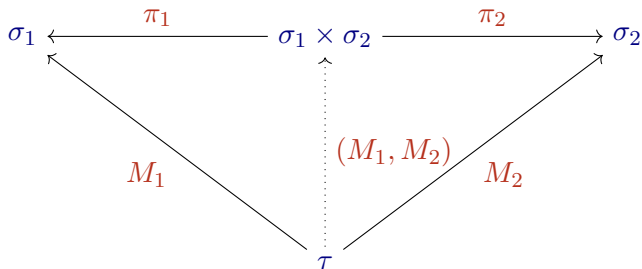
$$t_\sigma : \sigma \rightarrow 1$$

$$t_\sigma := \lambda x. \langle \rangle$$

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(ix) Binary products

We start with the objects and the arrows in the following diagram



where $(M_1, M_2) := \lambda x. \langle M_1 x, M_2 x \rangle : \tau \rightarrow \sigma_1 \times \sigma_2$,

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First, we prove that the diagram is commutative.

$$\begin{aligned}\pi_i \circ (M_1, M_2) &: \tau \rightarrow \sigma_i \\ \pi_i \circ (M_1, M_2) &= \pi_i \circ \lambda y. \langle M_1 y, M_2 y \rangle \\ &= \lambda x. \pi_i ((\lambda y. \langle M_1 y, M_2 y \rangle) x) \\ &= \lambda x. \pi_i \langle M_1 x, M_2 x \rangle \\ &= \lambda x. M_i x \\ &= M_i\end{aligned}$$

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Typed Lambda-Calculus

Next, we prove the uniqueness of arrow (M_1, M_2) . Let $N : \sigma_1 \times \sigma_2$ such that $\pi_i \circ N = M_i$. Then

$$\begin{aligned}(M_1, M_2) &= \lambda x. \langle M_1 x, M_2 x \rangle \\ &= \lambda x. \langle (\pi_1 \circ N) x, (\pi_2 \circ N) x \rangle \\ &= \lambda x. \langle (\lambda y. \pi_1 (N y)) x, (\lambda y. \pi_2 (N y)) x \rangle \\ &= \lambda x. \langle \pi_1 (N x), \pi_2 (N x) \rangle \\ &= \lambda x. N x \\ &= N\end{aligned}$$

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(x) Finitary products

Since **TLC** has binary products and a terminal object (nullary product) then it has finite products.

Outline

Typed Lambda-Calculus

References

References

- S. Abramsky and N. Tzevelekos (2011). Introduction to Categories and Categorical Logic. In: New Structures for Physics. Ed. by Bob Coecke. Vol. 813. Lecture Notes in Physics. Springer, pp. 3–94. DOI: [10.1007/978-3-642-12821-9_1](https://doi.org/10.1007/978-3-642-12821-9_1) (cit. on p. 2).